

TRIOSCYBER VIRTUAL GLOBAL INTERNSHIP - DIGITAL FORENSICS SYLLABUS

1-MONTH STRUCTURED TRAINING PROGRAM

Program Duration	1 Month
Live Classes	4 Sessions
Duration per Class	2 Hours
Total Live Training	8 Hours
Daily Learning Tasks	30 Tasks
Training Level	Beginner to Intermediate
Primary Focus	Digital Evidence, Forensic Acquisition, Disk, OS, Memory, Mobile, Application, Incident Response and Reporting

PROGRAM OVERVIEW

This syllabus is designed for a one-month Digital Forensics internship consisting of four live classes of two hours each. The program progresses from digital evidence and preservation to acquisition, disk and operating system forensics, memory, browser, email, mobile, application, incident response, timeline analysis and forensic reporting.

LEARNING OBJECTIVES

- Understand digital evidence, preservation, acquisition and forensic integrity.
- Analyze disk, file system, operating system, browser, email and application artifacts.
- Understand mobile device evidence and basic mobile forensic examination.
- Perform basic memory analysis and identify processes, sessions and network artifacts.
- Correlate forensic artifacts and reconstruct a sequence of digital events.
- Apply basic forensic investigation and incident response workflows.
- Document findings and prepare a structured forensic investigation report.

LIVE CLASS & MODULE DISTRIBUTION

Class	Modules	Main Focus
Class 1	1.1 - 1.4	Digital Evidence, Preservation, Acquisition, Disk and Operating System Forensics
Class 2	1.5 - 1.8	Memory, Browser, Email and Mobile Device Forensics
Class 3	1.9 - 1.12	Application and Cloud Artifacts, Malware Evidence, Incident Response and Anti-Forensics
Class 4	1.13 - 1.15	Timeline Analysis, Investigation, Evidence Correlation and Forensic Reporting

CLASS 1 - MODULES 1.1 TO 1.4

1.1 Digital Forensics Fundamentals

- Digital Evidence and Sources of Digital Evidence
- Types of Digital Evidence
- Digital Forensics Investigation Lifecycle
- Forensic Triage and Initial Assessment
- Crime Scene and Evidence Scene Management
- Identification of Relevant Evidence
- Order of Volatility
- Evidence Preservation

1.2 Evidence Acquisition and Preservation

- Evidence Handling Procedures
- Chain of Custody
- Digital Evidence Collection
- Forensic Imaging
- Live and Dead Acquisition
- Write Protection
- Evidence Hashing and Verification
- Evidence Integrity Validation

1.3 Disk and File System Forensics

- Disk and Storage Fundamentals
- Partitions and File Systems
- Files and Directory Structures
- File and Metadata Analysis
- File Signatures and File Types
- Deleted File Recovery
- Data Carving
- Unallocated and Slack Space
- Hidden Data and Partition Analysis
- Disk Image Examination

1.4 Operating System Forensics

- Windows Forensic Artifacts
- Windows Registry
- Windows Event Logs
- User and System Activity
- Program Execution Artifacts
- Linux Forensic Artifacts
- Linux Logs and Command History
- macOS Forensic Artifacts
- Operating System Activity Reconstruction

End-of-Class Practical Demonstration

Practical Objective - Demonstrate evidence identification, preservation, acquisition and basic disk and operating-system artifact examination using an authorized training evidence set.

CLASS 2 - MODULES 1.5 TO 1.8

1.5 Memory Forensics

- Volatile Memory and RAM
- Memory Acquisition Fundamentals
- Running Processes and Services
- Network Connections in Memory
- Loaded Modules and Libraries
- User Sessions and Logged-In Users
- Suspicious Process Identification
- Malware-Related Memory Artifacts
- Basic Memory Analysis

1.6 Browser and Web Forensics

- Web Browser Forensic Artifacts
- Browser History
- Cookies and Web Storage
- Browser Cache
- Download History
- Search History
- Saved Sessions and Tabs
- Web Activity Reconstruction

1.7 Email and Communication Forensics

- Email Evidence and Mailbox Structures
- Email Headers and Metadata
- Sender and Recipient Analysis
- Email Routing Information
- Email Attachments and Embedded Content
- Deleted and Archived Emails
- Email Application Artifacts
- Messaging and Communication Artifacts

1.8 Mobile Device Forensics

- Mobile Device Evidence Sources
- Android and iOS Forensic Fundamentals
- Mobile Device Preservation and Isolation
- Mobile Evidence Acquisition Fundamentals
- Logical and File-System Acquisition
- Mobile Application Artifacts
- Messages, Calls and Contacts
- Browser and Search History
- Photos, Videos and Metadata
- Location and GPS Artifacts
- Mobile Backups
- Deleted Mobile Data

End-of-Class Practical Demonstration

Practical Objective - Demonstrate basic memory, browser, email and mobile artifact examination and identify evidence relevant to a forensic investigation.

CLASS 3 - MODULES 1.9 TO 1.12

1.9 Application and Cloud Forensics

- Application Forensic Artifacts
- Application Logs and Configuration Data
- Document and Office File Artifacts
- PDF and Document Metadata
- Cloud-Synchronized Application Artifacts
- Cloud Account and Activity Evidence
- File Synchronization and Sharing Artifacts
- Application Activity Reconstruction

1.10 Malware and Suspicious File Forensics

- Malware-Related Digital Evidence
- Identification of Suspicious Files
- File Hashing and Known-File Identification
- Basic Static File Examination
- Basic Dynamic Analysis
- Persistence and Autorun Artifacts
- Malware-Related System Artifacts
- Suspicious File and Host Evidence Correlation

1.11 Incident Response and Forensic Investigation

- Incident Identification and Initial Triage
- Incident Scoping
- Evidence Identification and Preservation
- Collection of Volatile and Non-Volatile Evidence
- Host-Based Evidence Analysis
- Incident Timeline Reconstruction
- Indicators of Compromise Analysis
- Evidence Correlation
- Basic Attack Reconstruction
- Post-Incident Forensic Analysis

1.12 Anti-Forensics and Evidence Evasion

- Introduction to Anti-Forensics
- Data Wiping and Secure Deletion
- Log Tampering and Artifact Removal
- File and Metadata Manipulation
- Encryption and Obfuscation
- Steganography Detection
- Identifying Anti-Forensic Indicators
- Countering Anti-Forensic Techniques

End-of-Class Practical Demonstration

Practical Objective - Demonstrate application and cloud artifact review, suspicious-file examination, incident triage and recognition of anti-forensic indicators.

CLASS 4 - MODULES 1.13 TO 1.15

1.13 Digital Timeline and Artifact Analysis

- Forensic Timeline Creation
- File System Timestamps
- User Activity Reconstruction
- System Activity Reconstruction
- Application Activity Reconstruction
- Log and Artifact Analysis
- Artifact Correlation
- Cross-Source Evidence Correlation
- Reconstruction of Digital Events

1.14 Digital Forensic Investigation

- Case Preparation and Evidence Review
- Evidence Examination
- Identification of Relevant Artifacts
- Evidence Correlation Across Sources
- Investigation Timeline Development
- Suspicious Activity Analysis
- Reconstruction of Events
- Validation of Findings
- Case-Based Digital Forensic Investigation

1.15 Forensic Documentation and Reporting

- Forensic Examination Documentation
- Case Notes and Evidence Records
- Findings and Conclusions
- Evidence Validation and Quality Checks
- Supporting Screenshots and Exhibits
- Forensic Report Structure
- Presentation of Forensic Findings
- Incident Response Reporting
- Final Investigation Report

End-of-Class Practical Demonstration

Practical Objective - Demonstrate timeline creation, cross-source evidence correlation, event reconstruction and preparation of concise forensic finding.

RECOMMENDED 2-HOUR LIVE CLASS FORMAT

Time	Activity
0:00 - 0:15	Review previous tasks, address common questions and introduce the learning objectives.
0:15 - 1:00	Explain theory, concepts, terminology and forensic examples for the scheduled modules.
1:00 - 1:10	Short break and transition to the evidence examination.
1:10 - 1:50	Guided evidence examination using sample forensic datasets, disk images or authorized training environments.
1:50 - 2:00	Summarize key takeaways, explain the upcoming daily tasks and conduct Q&A.

DIGITAL FORENSIC INVESTIGATION WORKFLOW

- Identify and Scope Evidence
- Preserve Evidence and Maintain Chain of Custody
- Acquire and Verify Evidence
- Examine Forensic Artifacts
- Analyze and Correlate Evidence
- Reconstruct Digital Events
- Document Findings and Supporting Evidence
- Prepare the Forensic Report

IMPORTANT NOTE

All evidence examination activities should use authorized training datasets, isolated labs or systems for which appropriate permission exists.